Chapter 8 – Markov Chains (Part 1)

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1 Introduction

1. Markov Chains are special cases of time series with discrete state space satisfying the Markov Property.

2. Consider a sequence of discrete valued random variables $(X_0, X_1, \cdots, X_k)$. Note that X_0 is the initial state and it is possible for the sequence to go on forever.

3. For example, if we are interested in weather patterns categorised as S=sunny, R=rain and C=cloudy, we could have observed

$$SSCCRCSSSCRCC$$

and so on, where the initial zeroth day is $X_0 = S$, then $X_1 = S$, and 12 days after the zeroth day, $X_{12} = C$.

4. However in this case and in all cases, it is more convenient to code the states into integer values, for example, code $X_i = 1$ for sunny, $X_i = 2$ for cloudy and $X_i = 3$ for rainy. Then the sequence in item 1.3, can be rewritten as

$$1122321112322$$

5. Now suppose this weather pattern for this city is tracked over a very long period, and it is observed that for any day t,

$$P[X_{t+1} = x_{t+1} | X_t = x_t, X_{t-1} = x_{t-1}, \cdots, X_{t-k} = x_{t-k}]$$
$$= P[X_{t+1} = x_{t+1} | X_t = x_t],$$

that is, the probably that whatever state it is going to be tomorrow only depends on today and not on any other days prior to today, then we say that this discrete state space time series is a Markov Chain, or that it satisfies the Markov Property.

6. Suppose we have a Markov Chain with n states labelled $1, 2, \cdots, n$ and that $p_{ij} = P[X_{t+1} = j | X_t = i]$ is the one-step transition probability of starting from state i at any time t and going into state j one step (or one period) later, then the probabilities can be gathered into an $n \times n$ matrix P where the ij-th entry (that is, the entry in the ith row and jth column) is p_{ij}.

7. **Example 1**. Suppose our sunny, rainy and cloudy example has probabilities given by the one-step transition matrix P as

$$P = \begin{pmatrix} 0.4 & 0.3 & 0.3 \\ 0.3 & 0.4 & 0.3 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}.$$

Then this is how we interpret the matrix.

(a) The probabilities across the 1st row give the probabilities of starting from state 1 (sunny) at any time t and ending in the same state 1 (sunny) as 0.4, from state 1 going on to state 2 (cloudy) as 0.3, and from state 1 going on to state 3 (rainy) as 0.3, one period later at time $t+1$. In other words $p_{11} = 0.4$, $p_{12} = 0.3$ and $p_{13} = 0.3$.

(b) The probabilities across the 2nd row give the probabilities of starting from state 2 (cloudy) at any time t and ending in state 1 (sunny) as 0.3, from state 2 staying state 2 (cloudy) as 0.4, and from state 2 going on to state 3 (rainy) as 0.3, one period later at time $t+1$. In other words $p_{21} = 0.3$, $p_{22} = 0.4$ and $p_{23} = 0.3$.

(c) Likewise for the 3rd row, we have $p_{31} = 0.3$, $p_{32} = 0.5$ and $p_{33} = 0.2$.

Here is how we enter the matrix in **R** .

```
> P <- matrix(c(0.4,0.3,0.3,0.3,0.4,0.3,0.3,0.5,0.2),3,3,byrow=T)
> P
     [,1] [,2] [,3]
[1,]  0.4  0.3  0.3
[2,]  0.3  0.4  0.3
[3,]  0.3  0.5  0.2
```

8. Now the transition matrix P has a useful property. It can be use to calculate the probabilities of the states of the chain k periods ahead by simply looking at P^k.

9. **Example 2**. Use the transition matrix P from Example 1 and calculate P^2.

```
> P %*% P
     [,1] [,2] [,3]
[1,] 0.34 0.39 0.27
[2,] 0.33 0.40 0.27
[3,] 0.33 0.39 0.28
```

10. The entries in P^2 now give us the two-step transition matrix.

11. The ij-th entry in the ith row and jth column of this matrix is denoted by $p_{ij}^{(2)}$. Note that $p_{ij}^{(2)} \neq p_{ij}^2$.

12. The interpretation for the entry $p_{ij}^{(2)}$ is that

$$p_{ij}^{(2)} = P[X_{t+2} = j | X_t = i].$$

That is, the probability that we end up in state j two periods ahead given that we are currently in state i.

13. **Example 3**. Looking at items 1.3 and 1.4, where on the 12th day, it is cloudy. Suppose the day to day weather pattern follows the one-step transition matrix in Examples 1 and 2. How do we forecast the weather for day 13?

Since it is cloudy on day 12, we first create a row probability vector

$$x = (0 \quad 1 \quad 0).$$

This says that there is probability 1 that it is cloudy on day 12. (The second column corresponds to cloudy.)

```
> x <- cbind(0,1,0)
> x
     [,1] [,2] [,3]
[1,]    0    1    0
```

Now if we post multiply x by P, we get

$$xP = (0.3 \quad 0.4 \quad 0.3).$$

```
> x %*% P
     [,1] [,2] [,3]
[1,]  0.3  0.4  0.3
```

This tells us that on day 13, given that it is cloudy on day 12, the probability of day 13 being sunny is 0.3, the probability of day 13 remaining cloudy is 0.4, and the probability of day 13 being rainy is 0.3.

14. **Example 4**. Looking at items 1.3 and 1.4, where on the 12th day, it is cloudy. Suppose the day to day weather pattern follows the one-step transition matrix in Examples 1 and 2. How do we forecast the weather for day 14?

Since it is cloudy on day 12, we first create a row probability vector

$$x = (0 \;\; 1 \;\; 0).$$

We already have the object x stored in **R** since we created it for Example 3 above. Now since we want to calculate the probabilities of the weather 2 days ahead, we post multiply x by P^2, the two-step transition matrix.

```
> x %*% P %*% P
     [,1] [,2] [,3]
[1,] 0.33  0.4 0.27
```

This gives us

$$xP^2 = xPP = (0.33 \; 0.4 \; 0.27),$$

which tells that on day 14, given that it is cloudy on day 12, the probability of day 14 being sunny is 0.33, the probability of day 14 remaining cloudy is 0.4, and the probability of day 14 being rainy is 0.27.

2 Steady-State Probabilities

1. Sometimes a Markov Chain may settle into a steady-state condition.

2. Mathematical this means that the n step transition matrix P^n settles into some matrix Π that never changes as we keep on post multiplying again by P, that is, we formally write this as

$$\lim_{n\to\infty} P^n = \Pi.$$

3. However, in practice, this means that we keep multiplying P with itself over and over again until the result does not appear to change any more.

4. **Example 5**. Using the same one-step transition matrix in our weather examples in Section 1, we first compute some higher powers of P.

```
> P %*% P  %*% P
      [,1]  [,2]  [,3]
[1,] 0.334 0.393 0.273
[2,] 0.333 0.394 0.273
[3,] 0.333 0.395 0.272
```

and so on until P^7 and P^8 in this example.

```
> P %*% P %*% P %*% P %*% P %*% P %*% P
          [,1]      [,2]      [,3]
[1,] 0.3333334 0.3939393 0.2727273
[2,] 0.3333333 0.3939394 0.2727273
[3,] 0.3333333 0.3939395 0.2727272
> P %*% P %*% P %*% P %*% P %*% P %*% P %*% P
          [,1]      [,2]      [,3]
[1,] 0.3333333 0.3939394 0.2727273
[2,] 0.3333333 0.3939394 0.2727273
[3,] 0.3333333 0.3939394 0.2727273
```

Thus we see that after 8 days, the weather appears to settle into some predictable pattern that stays the same. This means that for any typical day thereafter, there is a probability of 0.3333333 that it will be sunny, 0.3939394 that it will be cloudy and 0.2727273 that it will be rainy. So we can write

$$\Pi = \begin{pmatrix} 0.3333333 & 0.3939394 & 0.2727273 \\ 0.3333333 & 0.3939394 & 0.2727273 \\ 0.3333333 & 0.3939394 & 0.2727273 \end{pmatrix}.$$

Furthermore, if we take $x\Pi$ whether $x = (1\ 0\ 0)$, $x = (0\ 1\ 0)$ or $x = (0\ 0\ 1)$, we get

$$x\Pi = (0.3333334\ 0.3939393\ 0.2727273),$$

which basically says the same thing: in the long term, there is a probability of 0.3333333 that it will be sunny, 0.3939394 that it will be cloudy and 0.2727273 that it will be rainy.

3 Further Examples

1. These are further examples using the one-step transition matrix

$$P = \begin{pmatrix} 0.4 & 0.3 & 0.3 \\ 0.3 & 0.4 & 0.3 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}$$

   in Example 1.

2. **Example 6**. Given that today is a sunny day, what is the probability that it is sunny tomorrow and cloudy the day after tomorrow?

   The answer is given by $p_{11} \times p_{12} = 0.4 \times 0.3 = 0.12$. Note that this is not the entry $p_{12}^{(2)}$ in the matrix P^2 because

$$p_{12}^{(2)} = p_{11} \times p_{12} + p_{12} \times p_{22} + p_{13} \times p_{32}.$$

3. **Example 7**. Given that today is cloudy, what is the probability that it is cloudy tomorrow, then rain the day after, and then cloudy again after the rainy day?

   We are looking at $C \to C \to R \to C$, or $2 \to 2 \to 3 \to 2$. So this is

$$p_{22} \times p_{23} \times p_{32} = 0.4 \times 0.3 \times 0.5 = 0.06.$$

4. The next example is an actuarial science example. Insurance companies are able to tabulate the year to year transition probabilities of their state of health of their members based on gender and their age range. These can be use to calculate for example the probabilities of a healthy member dying within a certain age, or a healthy member stricken by a critical illness by a certain age. The sick state is a state of illness which is recoverable. A critical illness is one that is not recoverable and the individual either stays in that state for years or the individual dies.

5. **Example 8**. Suppose the following year to year transition probabilities apply for a 60 year old female up to age 70.

	Healthy	Sick	Critically Ill	Dead
Healthy	0.8	0.1	0.05	0.05
Sick	0.6	0.15	0.15	0.1
Critically Ill	0	0	0.6	0.4
Dead	0	0	0	1

So take for example the first line of the table. This gives the probabilities of a healthy 60 year old female remaining healthy, turning sick, becoming critically or dead respectively by her 61st birthday. The same probabilities also apply if she is healthy right immediately after her 61st birthday and going into those respective states by her 62nd birthday.

(a) Calculate the probabilities of the state of health of a currently healthy 60 year old female by the day of her 62nd birthday.

We first code in the probabilities into a transition matrix P in **R** , and then calculate P^2.

```
> P <- matrix(c(0.8, 0.1, 0.05, 0.05,
+ 0.6,0.15,0.15,0.1,0, 0,0.6,0.4,0,0,0,1),4,4,byrow=T)
> P
     [,1] [,2] [,3] [,4]
[1,]  0.8 0.10 0.05 0.05
[2,]  0.6 0.15 0.15 0.10
[3,]  0.0 0.00 0.60 0.40
[4,]  0.0 0.00 0.00 1.00
> P %*% P
     [,1]   [,2]   [,3]  [,4]
[1,] 0.70 0.0950 0.0850 0.120
[2,] 0.57 0.0825 0.1425 0.205
[3,] 0.00 0.0000 0.3600 0.640
[4,] 0.00 0.0000 0.0000 1.000
```

From the first row of P^2, we see that she will still be healthy with probability 0.7, sick with probability 0.095, critically ill with probability 0.085, and dead with probability 0.12.

(b) What is the probability that this initially healthy 60 year old female will die sometime in her 63rd year, that is, sometime after her 62nd birthday but by or on her 63rd birthday?

We need to calculate P^3 first.

```
> P %*% P %*% P
       [,1]     [,2]     [,3]    [,4]
[1,] 0.6170 0.084250 0.100250 0.19850
[2,] 0.5055 0.069375 0.126375 0.29875
[3,] 0.0000 0.000000 0.216000 0.78400
[4,] 0.0000 0.000000 0.000000 1.00000
```

So the probability that she will be dead by or on her 63rd birthday is $p^{(3)}_{HD} = 0.1985$, the 14-th entry in the P^3 matrix. The probability that she will be dead by or on her 62nd birthday is $p^{(2)}_{HD} = 0.12$, the 14-th entry in the P^2 matrix. Therefore the probability that she dies during her 63rd year is

$$p^{(3)}_{HD} - p^{(2)}_{HD} = 0.1985 - 0.12 = 0.0785.$$

6. Here is an interesting observation from the previous Example 8. If I keep multiplying P with itself many, many times over, I get

```
            [,1]         [,2]         [,3]      [,4]
[1,] 0.001310727 0.0001790682 3.276816e-04 0.9981825
[2,] 0.001074409 0.0001467831 2.686023e-04 0.9985102
[3,] 0.000000000 0.0000000000 2.909813e-12 1.0000000
[4,] 0.000000000 0.0000000000 0.000000e+00 1.0000000
```

I continue again multiplying the result with P again many, many more times and I get

```
              [,1]         [,2]         [,3]      [,4]
[1,] 1.108579e-05 1.514513e-06 2.771447e-06 0.9999846
[2,] 9.087079e-06 1.241454e-06 2.271770e-06 0.9999874
[3,] 0.000000e+00 0.000000e+00 1.080470e-20 1.0000000
[4,] 0.000000e+00 0.000000e+00 0.000000e+00 1.0000000
```

You can see from the pattern that eventually the entries in first 3 columns are going to be 0 except for the entries in the last column, which is the dead state, all going to 1. So no matter what state we begin with, we will eventually end up with the dead state with probability 1. Also the dead state is an "absorbing state" since looking back at the original one-step transition matrix P in the bottom most row, the dead state can only go into the dead state with probability 1.